

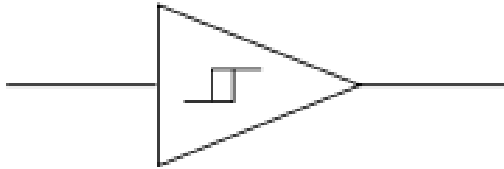
ANALOG & DIGITAL CIRCUITS

Lecture 7

Computer Science Department, TUCN

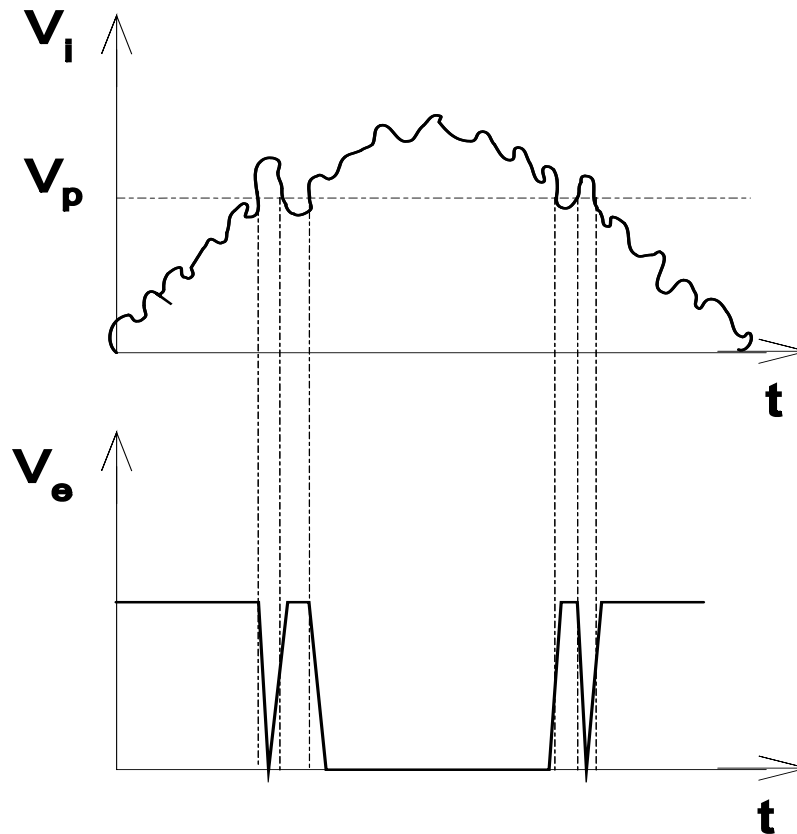
DIGITAL DEVICES

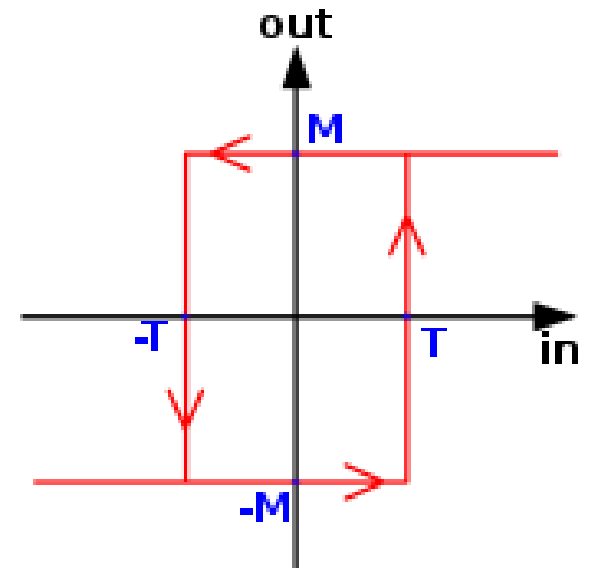
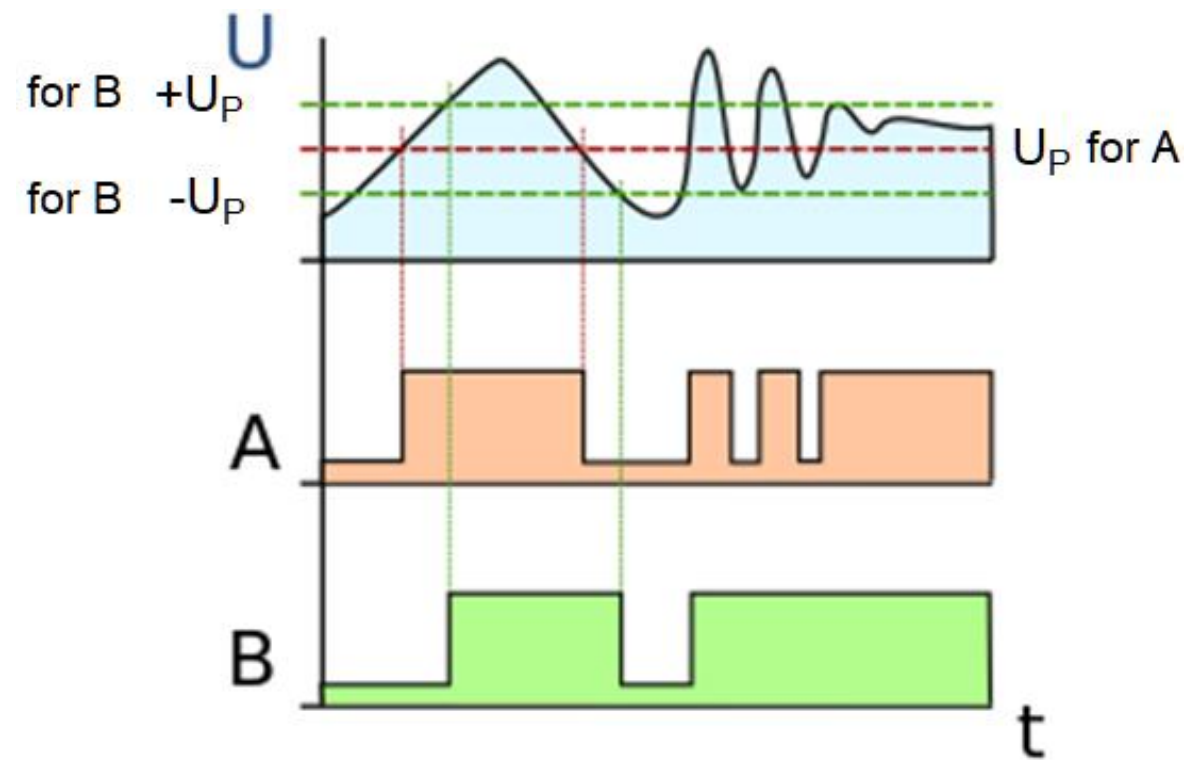
Schmitt Trigger (ST) Circuits



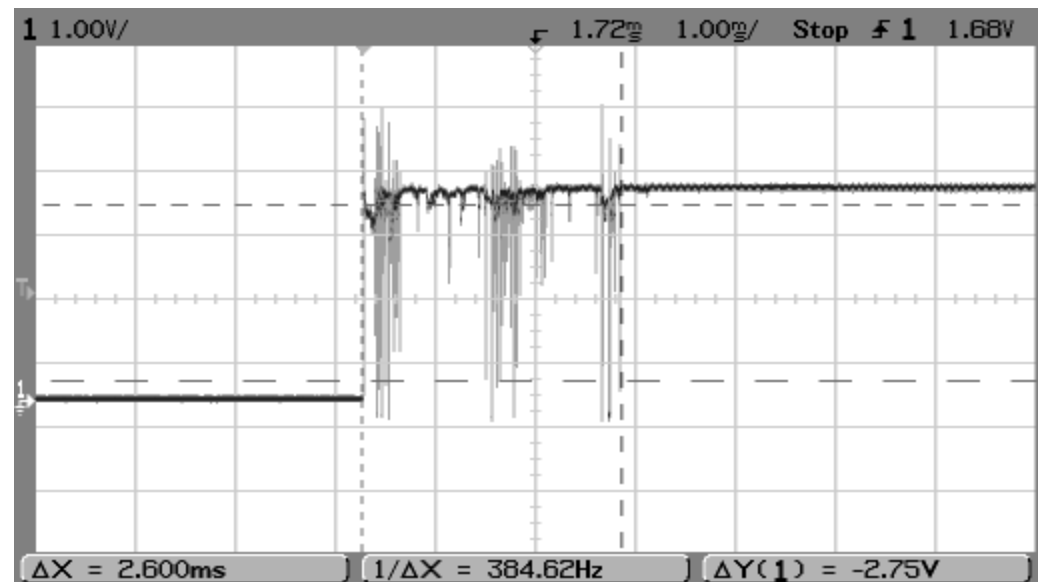
Schmitt triggers (pulse generators)

- Used for adapting the logical levels and for generating steep (firm) pulse edges
- Use: if at the inputs of some gates are applied signals with slow variation, together with disturbing (false) pulses: when connecting, in a numerical system the electromechanical parts (relays) with the logical ones, made of TTL or CMOS gates



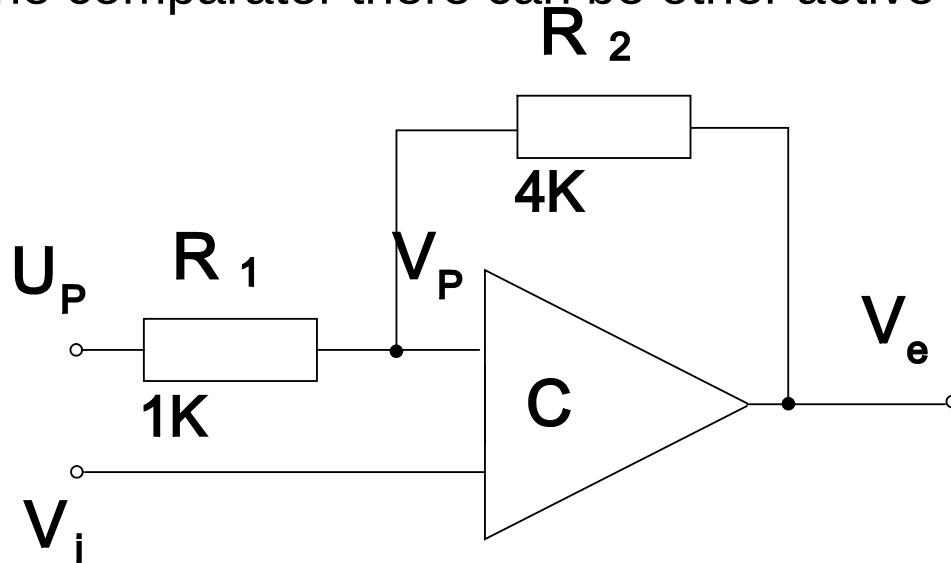


Switch contact bounce
seen on an oscilloscope

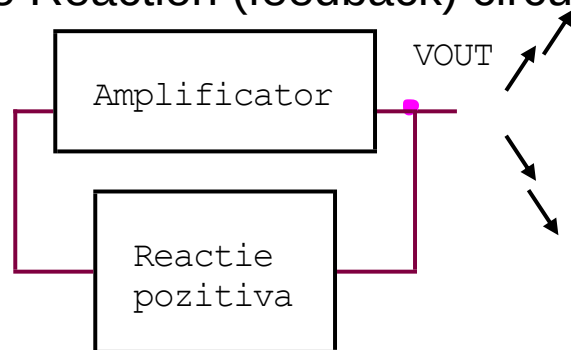


Schmitt triggers (pulse generators)

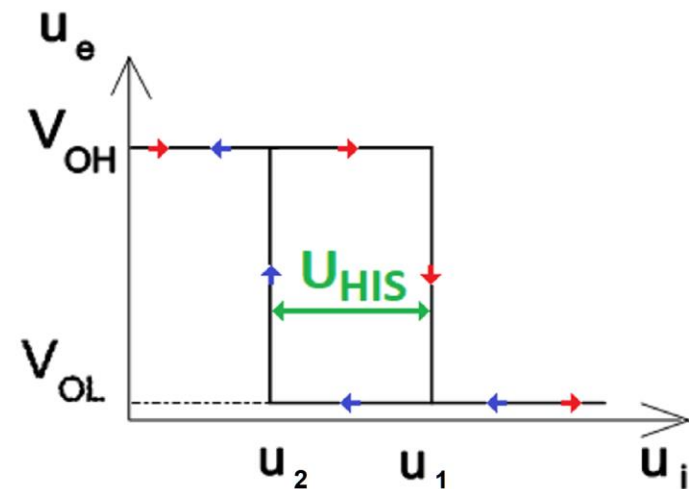
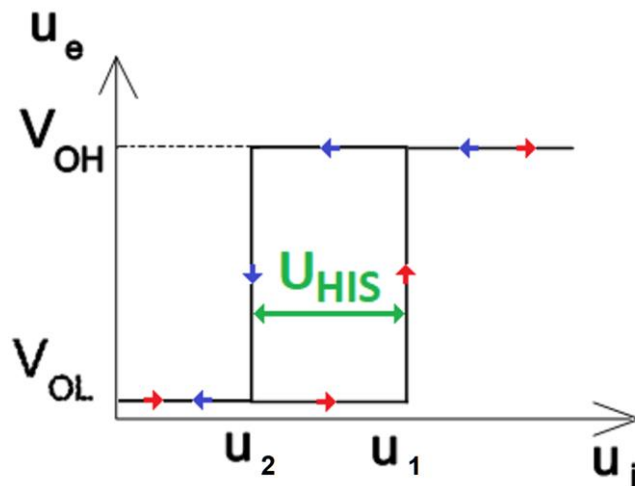
- Fundamental layout of a Schmitt trigger consists of a comparator with a positive reaction loop, made of two resistors
- Instead of the comparator there can be other active elements



Remember Positive Reaction (feedback) circuits

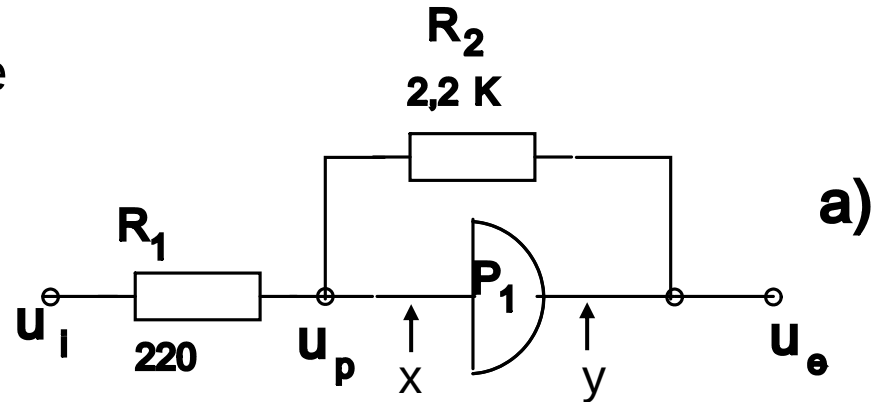


- The transfer characteristic: it presents two states, being a hysteresis characteristic
- Two positive consequences:
 - the transition between the two states is fast, even if the input signal has a slow variation
 - as long as the noise signal amplitude, which affects the input, is smaller than the hysteresis, there will be only one transition for the rising front of the input and only one transition for the falling front of the input signal

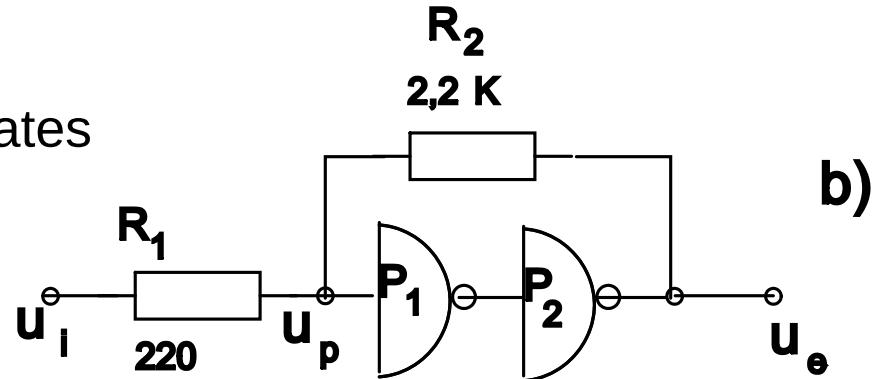


Schmitt Triggers realized with TTL gates

Schmitt trigger made of AND gate



Schmitt trigger made of NAND gates



- Consider negligible values of the input and output currents of used TTL gates, in comparison with the current that passes through the R1 and R2 resistors
- Voltage in point P, denoted by U_p

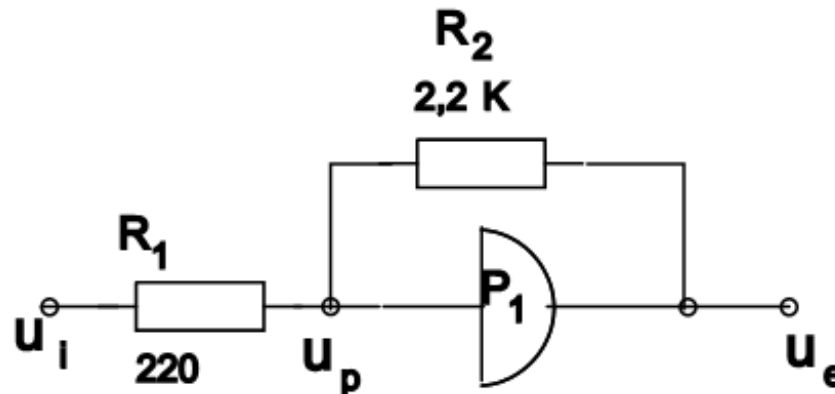
$$U_p = U_i - \frac{R_1}{R_1 + R_2}(U_i - U_e)$$

- For input voltage U_i of 0V, the output voltage is null as well:

$U_i = 0V \rightarrow U_e = 0V \Rightarrow$ non inverting gate

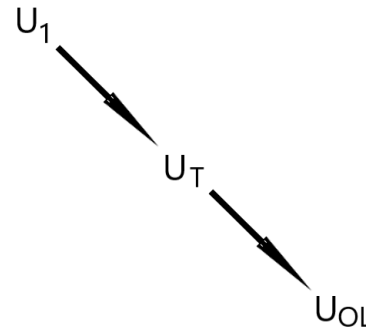
- If U_i increases, U_p increases too, and there will be U_p equal to threshold voltage of TTL gate U_T
- Consider $U_p = U_T$, $U_i = U_1$ and $U_e = U_{OL}$:

$$U_T = U_1 - \frac{R_1}{R_1 + R_2}(U_1 - U_{OL})$$

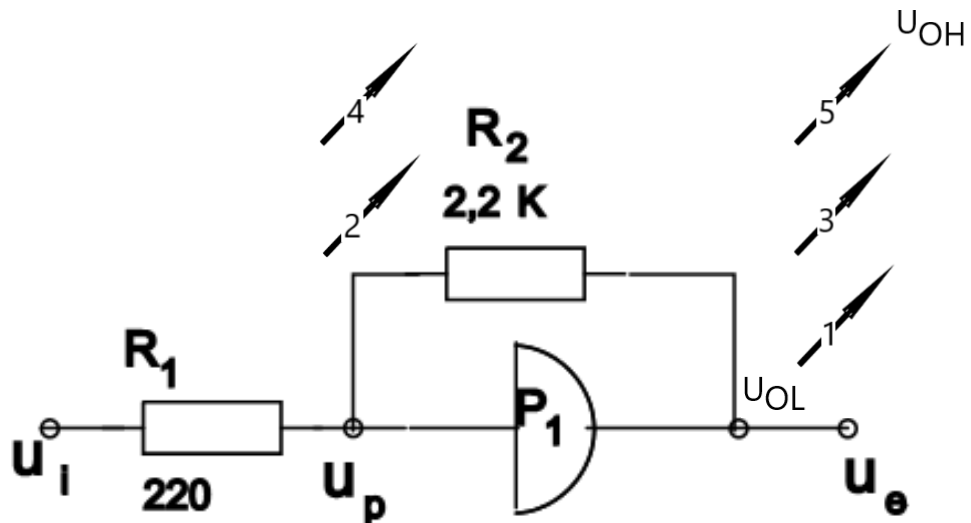
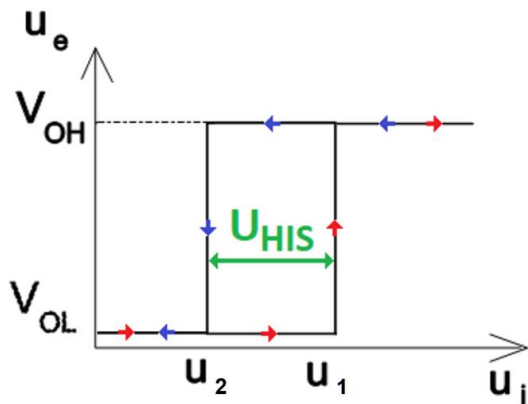


- Corresponding input voltage is (when U_p equals the threshold level):

$$U_1 = U_T \frac{(R_1 + R_2)}{R_2} - U_{OL} \frac{R_1}{R_2}$$

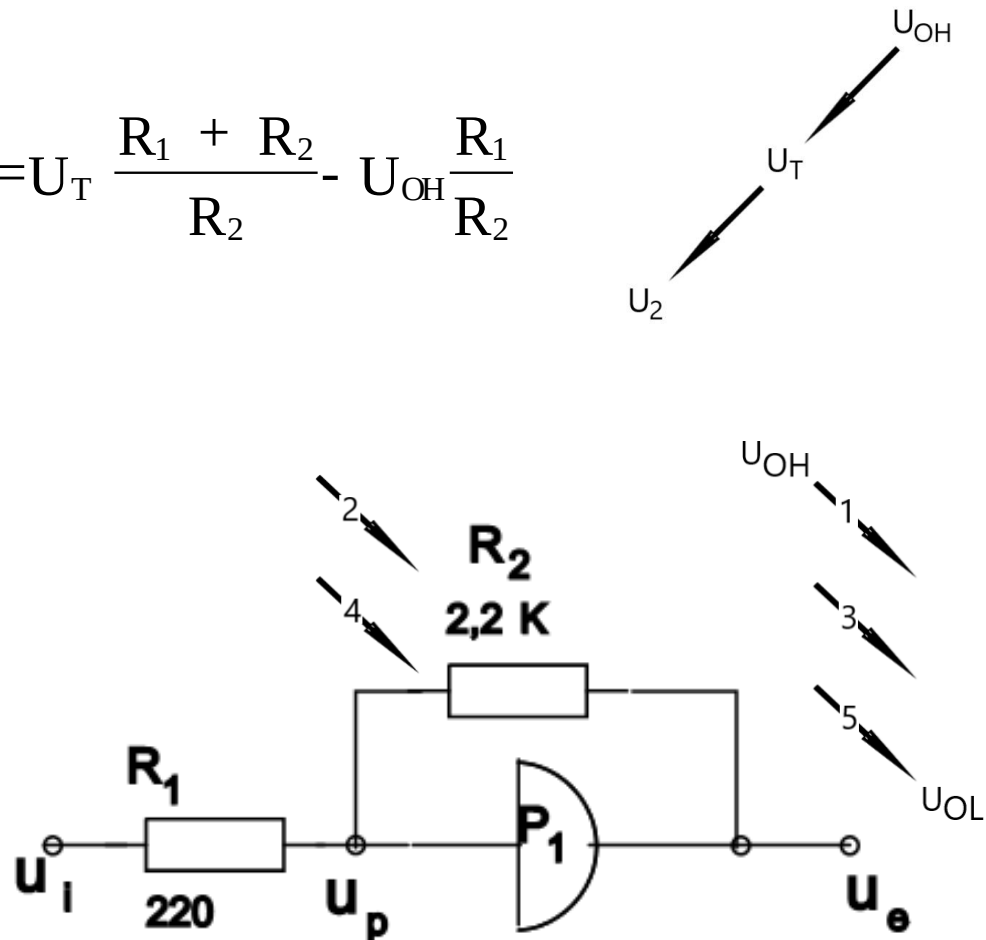
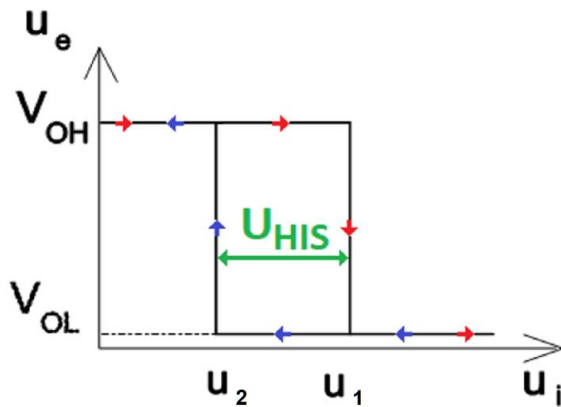


- When the input voltage exceeds the level U_1 , at which $U_p = U_T$, the switching of the gate takes place.
- The output voltage grows producing a positive reaction through the resistor R_2 and passing rapidly on high logical level, U_e taking the value U_{OH} regardless of the continuous increase of the input voltage



- When decreasing the input voltage, phenomenon is similar:
 - when reaching (in point P) the threshold voltage of the TTL gate, the output voltage begins dropping due to the positive reaction through the R_2 resistor, going fast to low logical level, U_{OL} .
 - the input voltage that determines the switching is calculated similarly:

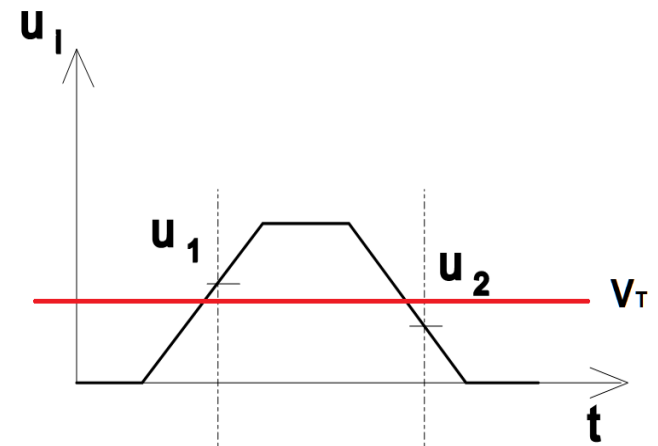
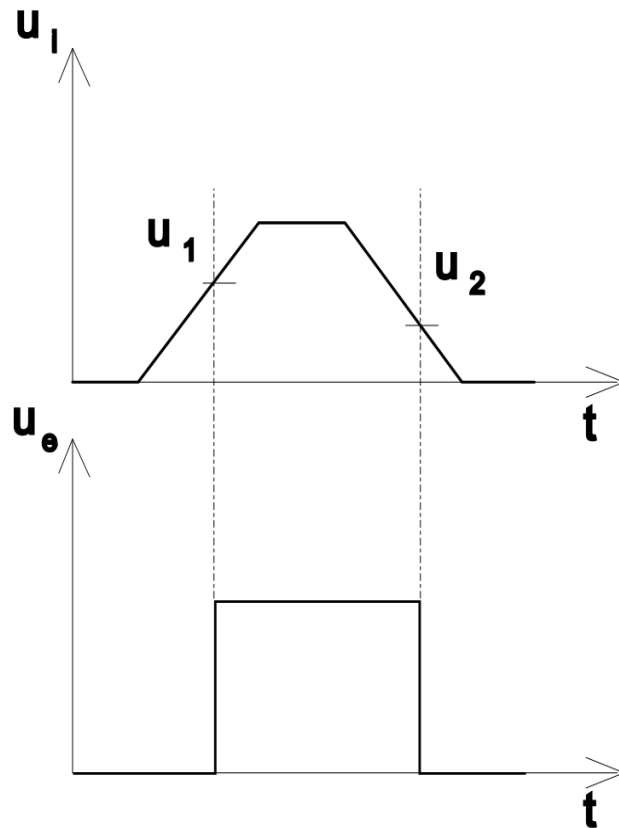
$$U_2 = U_T \frac{R_1 + R_2}{R_2} - U_{OH} \frac{R_1}{R_2}$$

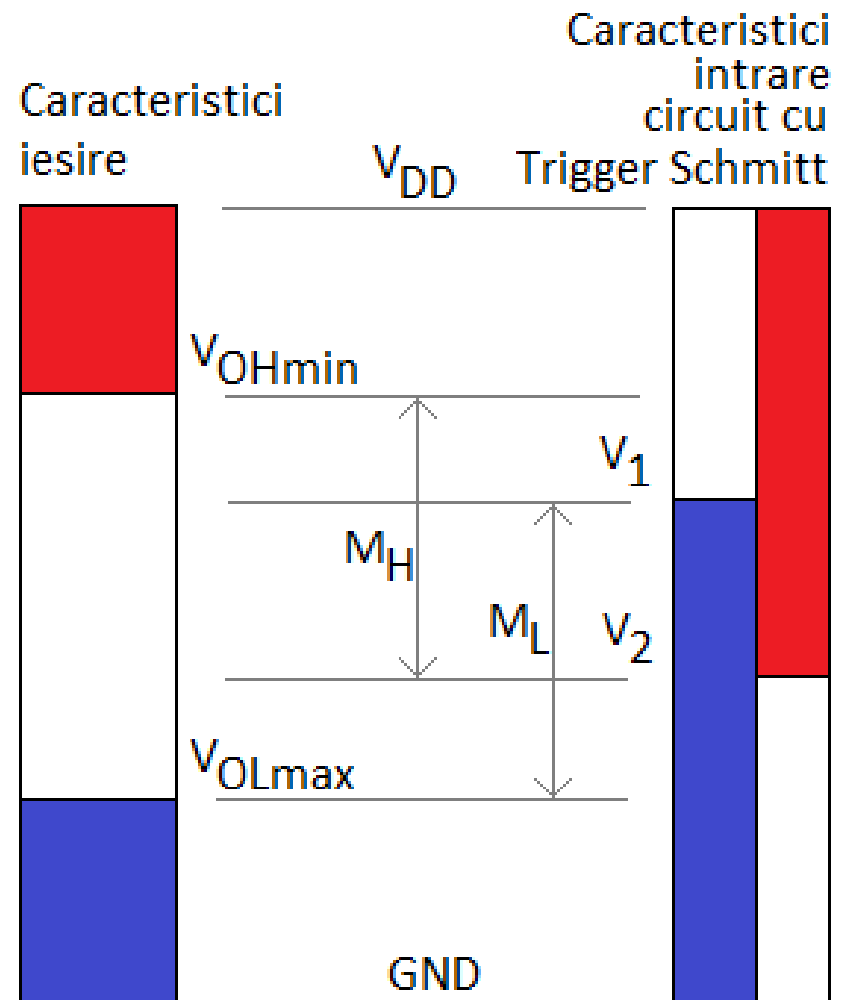
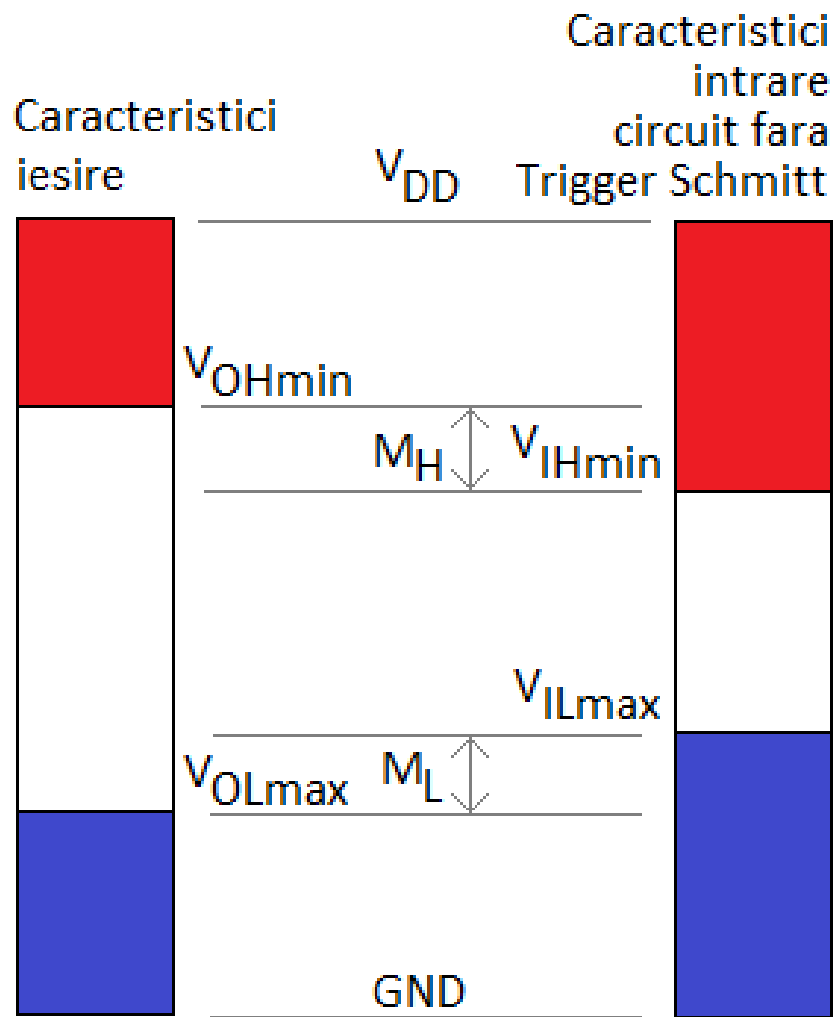


- It results the hysteresis voltage:

$$U_{HIS} = U_1 - U_2 = \frac{R_1}{R_2}(U_{OH} - U_{OL})$$

- Modifying the ratio between R_1 and R_2 the hysteresis cycle can be modified, noting that R_1 has the upper boundary at the value of 390 ohms

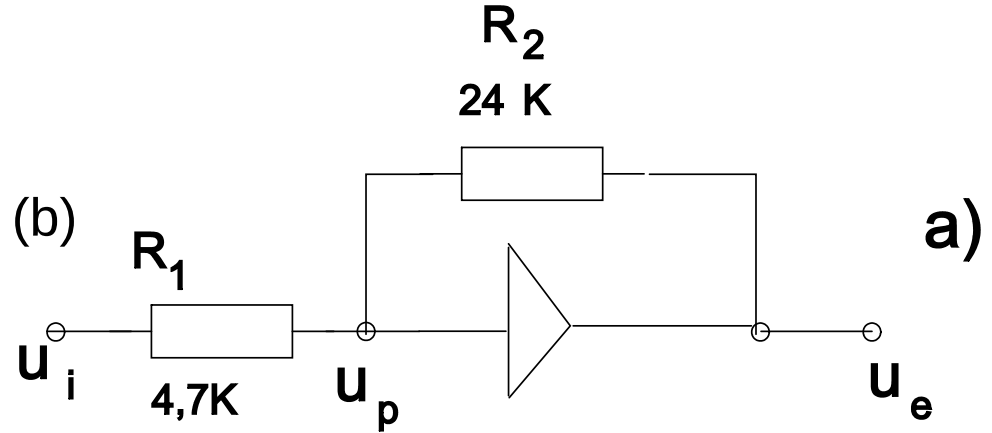




- The circuit layout shows identical input/output circuits as for every TTL gate; the trigger itself is made of T_2 and T_3 transistors.
- The electronic layout of the integrated Schmitt trigger contains compensation circuits for avoiding temperature influence, the threshold voltages being set in the prescribed temperature and supply voltage range.
- The switching thresholds for a supply voltage $V_{CC}=5V$ are:
 - $U_1=1,7V$ and $U_2=0,9V$, determining a hysteresis voltage of $0,8V$
- Other typical values for the circuit: $I_{IL} = -1,6mA$, $I_{IH} = 40\mu A$, $V_{ILmax}=0,7V$ and $V_{IHmin}=2V$.
- The maximum propagation delays for the CDB413 integrated circuit are: $t_{pHL} = 30ns$ and $t_{pLH} = 35 ns$.

Schmitt triggers made of CMOS gates

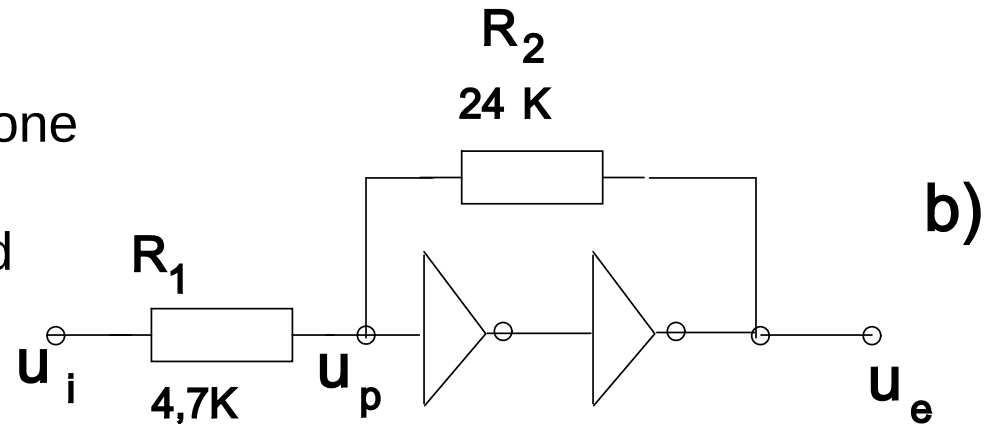
- Schmitt trigger can be obtained either with non-inverter CMOS circuits (a) or with inverter circuits (b)



- The functioning of the trigger with CMOS gates is identical with the one with TTL gates;
- Calculus formulas of the threshold voltages are directly obtained:

$$U_1 = U_T \cdot \frac{R_1 + R_2}{R_2} - U_L \cdot \frac{R_1}{R_2}$$

$$U_2 = U_T \cdot \frac{R_1 + R_2}{R_2} - U_H \cdot \frac{R_1}{R_2}$$



If we consider $U_L = 0V$, $U_H = V_{DD}$ and $U_T = V_{DD}/2$, we obtain:

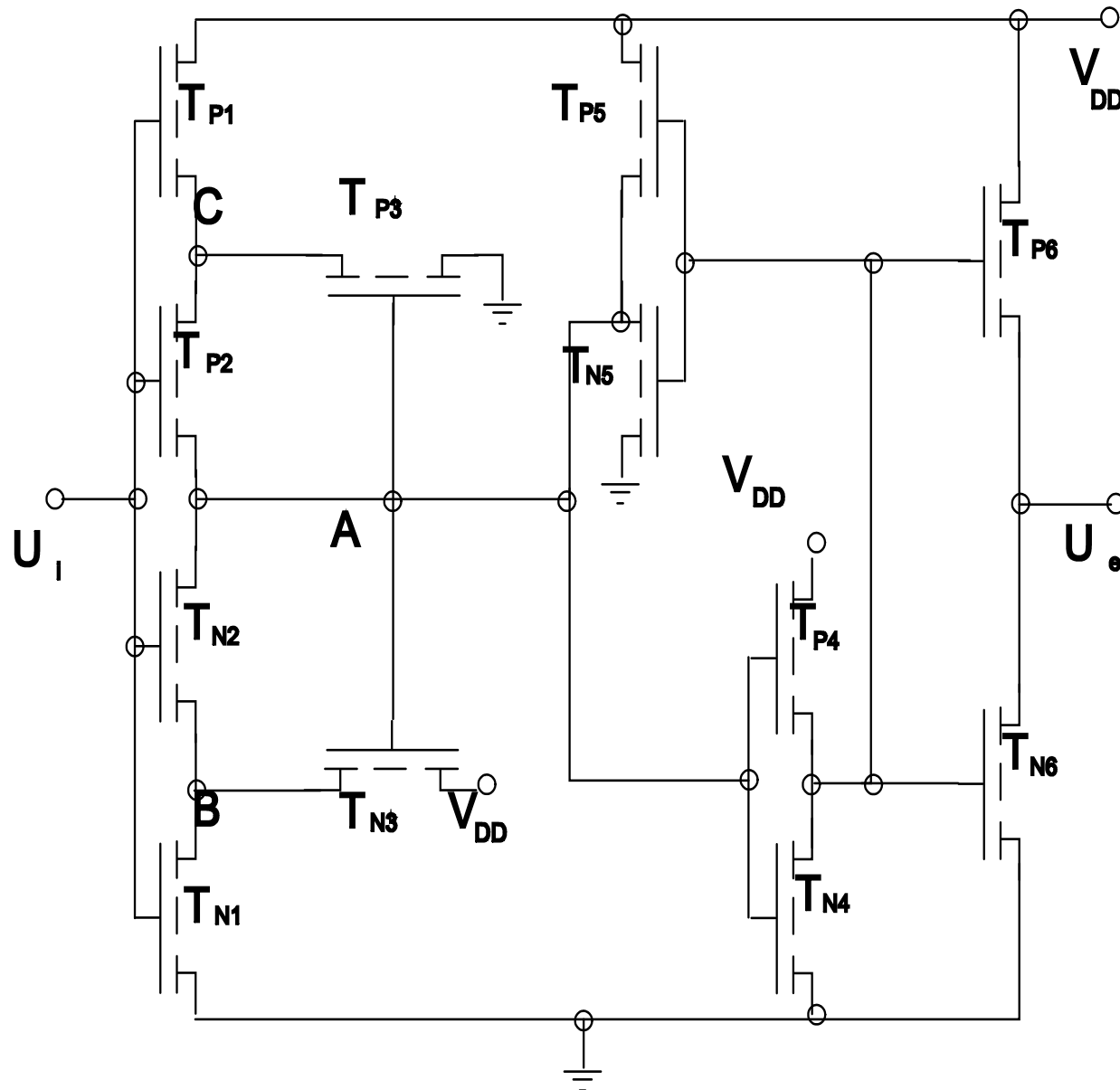
$$U_1 = \frac{R_1 + R_2}{2R_2} \cdot V_{DD}$$

$$U_2 = \frac{R_2 - R_1}{2R_2} \cdot V_{DD}$$

Results a hysteresis voltage:

$$U_{HIS} = U_1 - U_2 = \frac{R_1}{R_2} \cdot V_{DD}$$

Integrated CMOS Schmitt trigger circuits

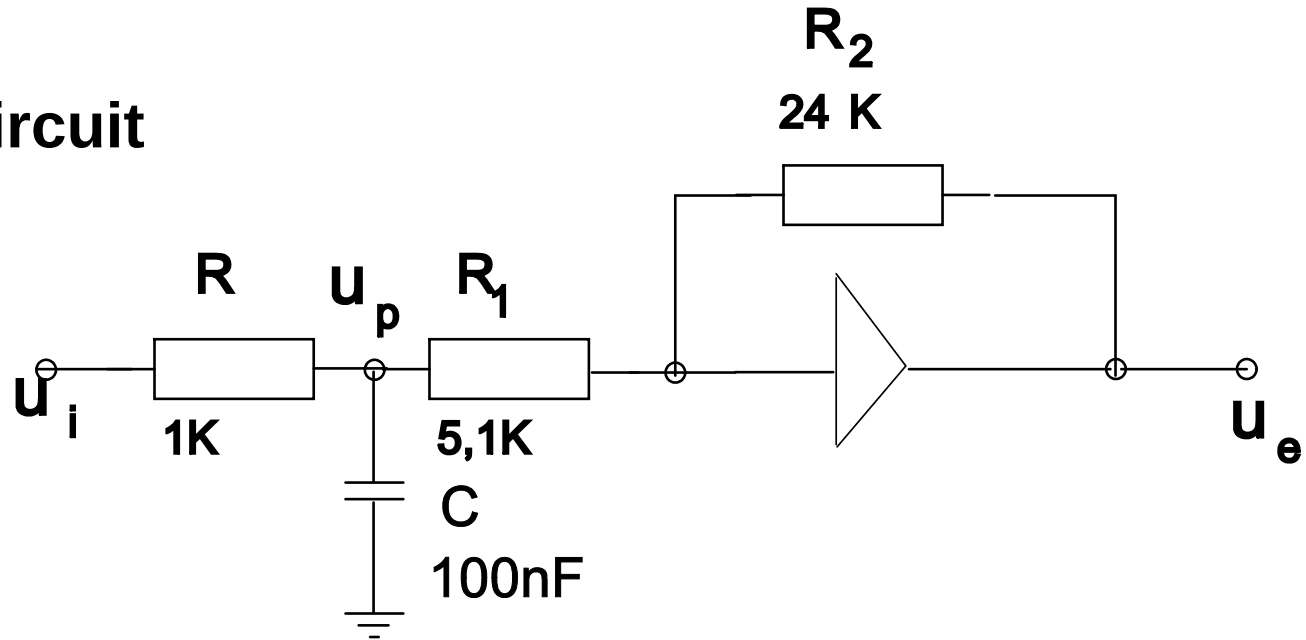


Layout
integrated circuit
MMC 4093

- Figure behind presents the electrical diagram of an integrated Schmitt trigger with CMOS transistors.
- The input circuit is made of six transistors, three with inductive channel **p** and three with inductive channel **n**.
- The two CMOS inverters obtained through the pairs T_{P4} , T_{N4} , respectively T_{P5} , T_{N5} form a RS type circuit (latch) for stabilization and detainment of the voltage in point A.
- The output circuit is obtained from the CMOS inverter with the transistors T_{P6} and T_{N6} .
- The transfer characteristics of a TS integrated CMOS depends on the voltage supplies.
 - For $V_{DD}=5V$, the integrated circuit presents a typical hysteresis voltage $U_{HIS}=0,9V$, and for a supply of $V_{DD}=10V$, it presents $U_{HIS}=2,3V$

Applications of the Schmitt trigger

Delay Circuit



- Delay circuit obtained with a TS and a low-pass RC circuit
- For joining circuits, necessary respecting the condition:
$$(R_1 + R_2) > 10 \cdot R$$
- This way the time constant of the integrating circuit stays $\tau = RC$, and the hysteresis voltage of the ST will not be influenced by the resistor R

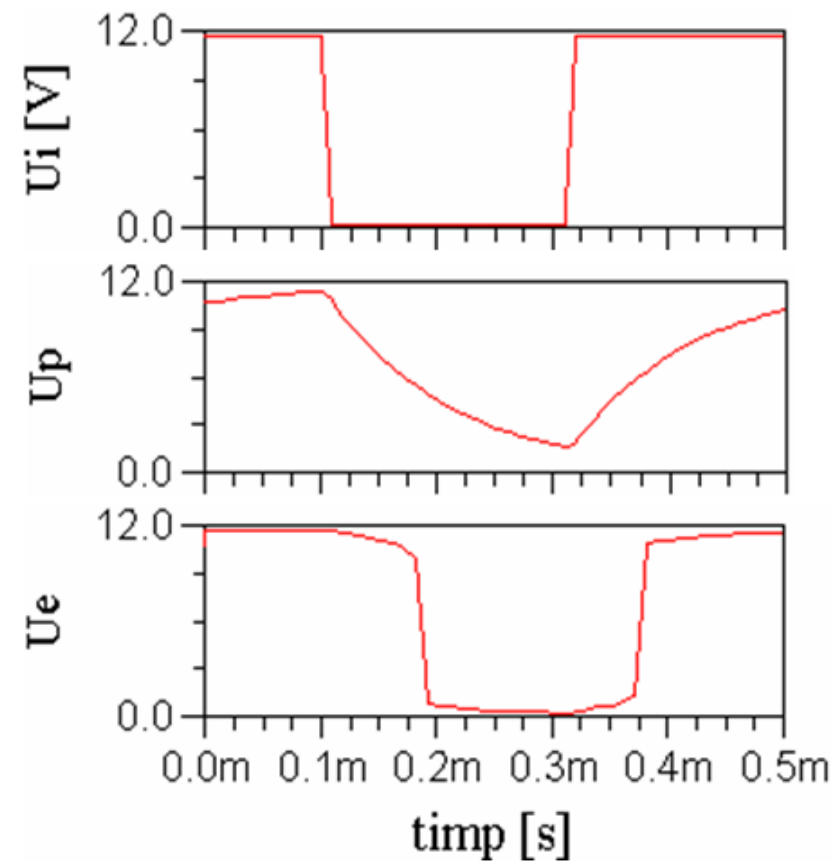
- By applying at the input a pulse signal, at the output the signal will be delayed, with δt_1 for the forefront and respectively δt_2 for the back front.
- The voltages U_1 and U_2 being the threshold voltages of the TS, the relations for the two delays are:

$$\delta t_1 = RC \cdot \ln \frac{V_{DD}}{V_{DD} - U_1}$$

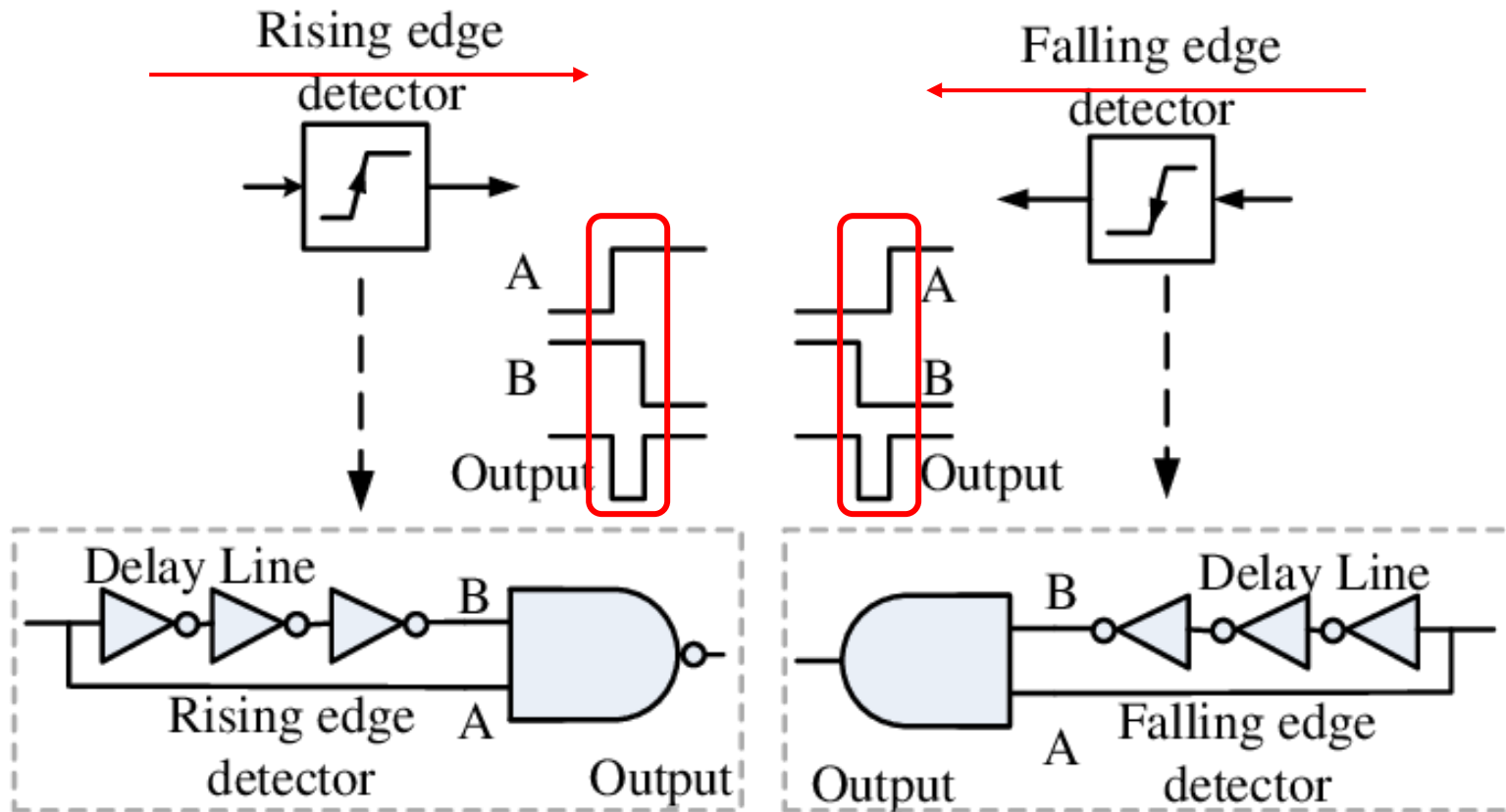
$$\delta t_2 = RC \cdot \ln \frac{V_{DD}}{U_2}$$

For $V_T = V_{DD}/2$:

$$\delta t_1 = \delta t_2 = RC \cdot \ln \frac{2 \cdot R_2}{R_2 - R_1}$$

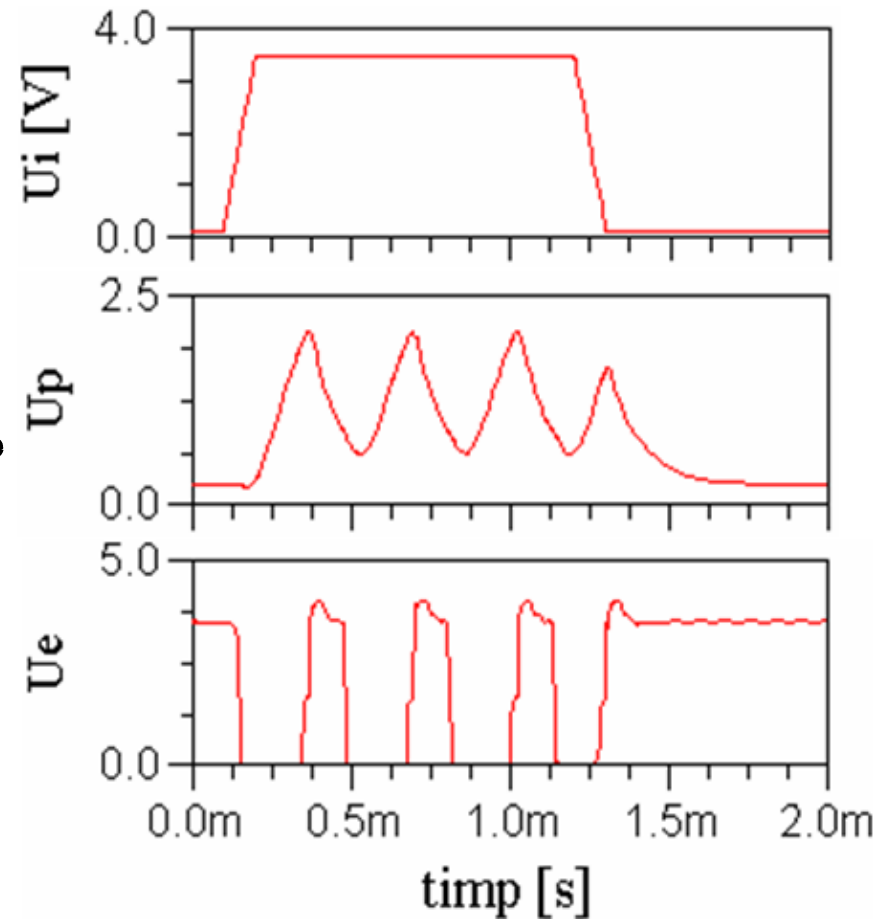
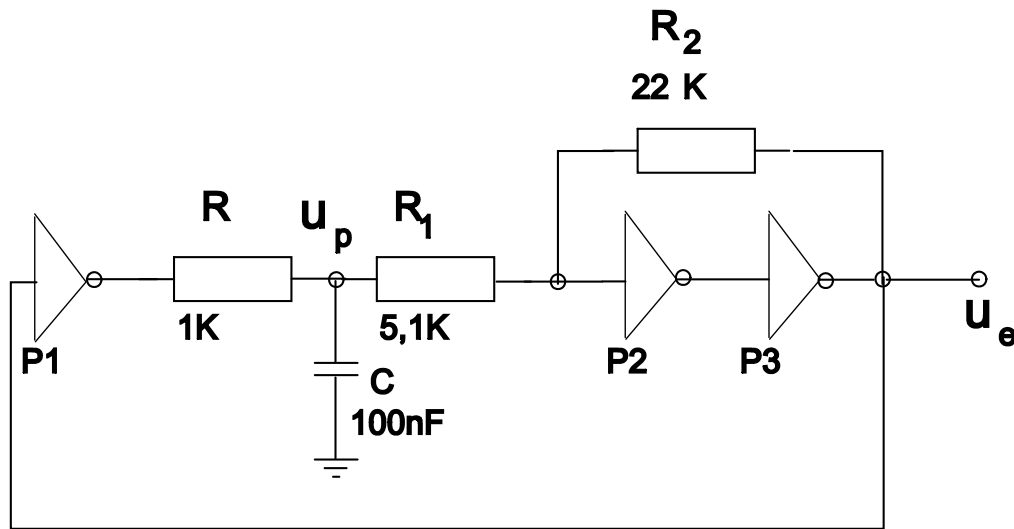


Edge detector circuits



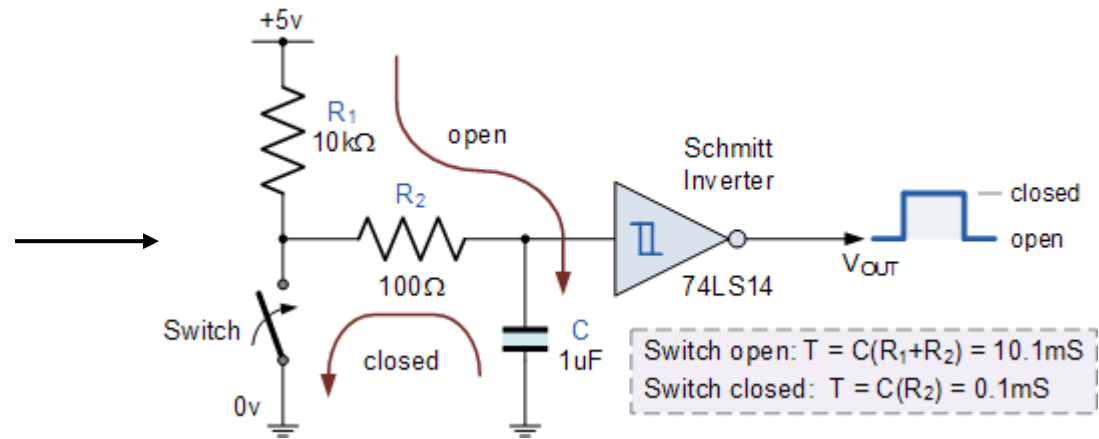
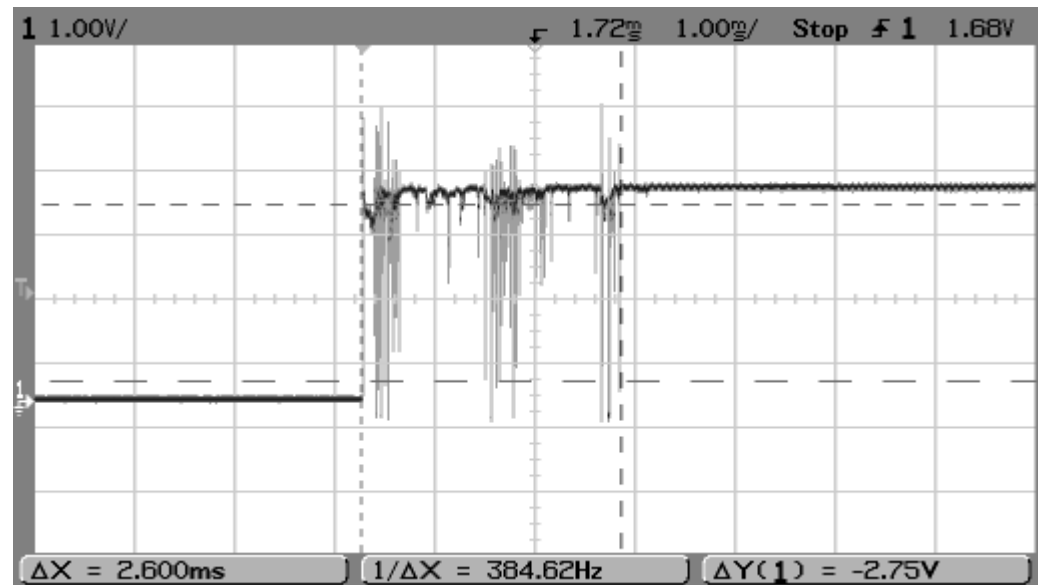
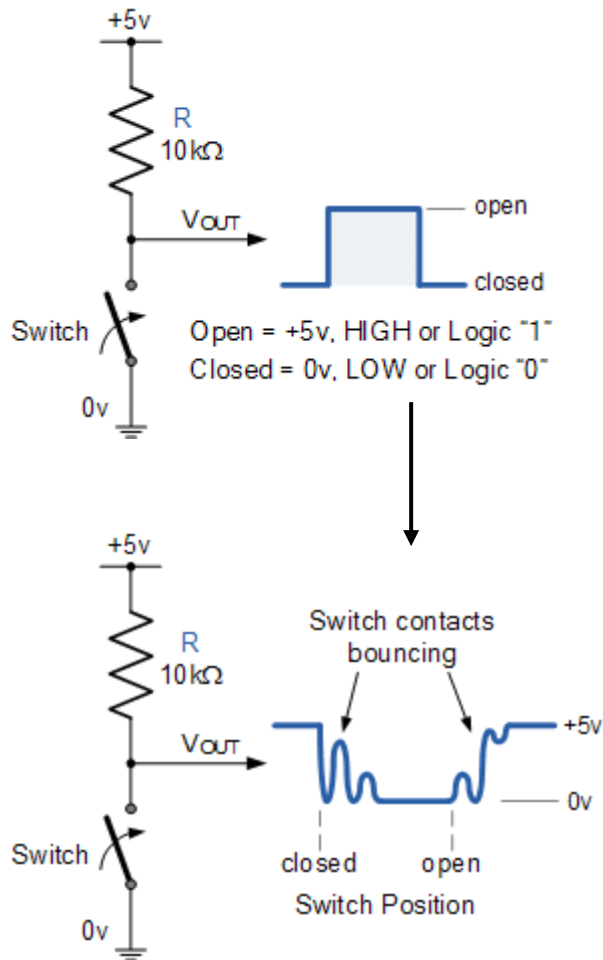
- NAND gate -> the circuit will detect the positive edges;
- NOR gate -> the circuit will detect the negative edges
- XOR gate -> the circuit will detect both edges

Edge detector circuits using TS



- Layout behind represents an edge detector circuit, obtained with TS circuit and RC low-pass circuit
- If at the output of the circuit from the figure we put a NAND gate, the circuit will detect the positive edges;
- if we put a NOR gate, the detection of the negative edges will be realized, and the XOR gate will allow the detection of both edges

Switch contact/switch bounce
seen on an oscilloscope →



Proposed Problems

- Design a delay circuit with a TS circuit using CMOS gates, to provide a delay for the input signal of 0,1 ms. CMOS gates use a power supply source of 5V.

Assuming:

$$R=1K\Omega, R_1=5,1K\Omega, R_2=24K\Omega$$

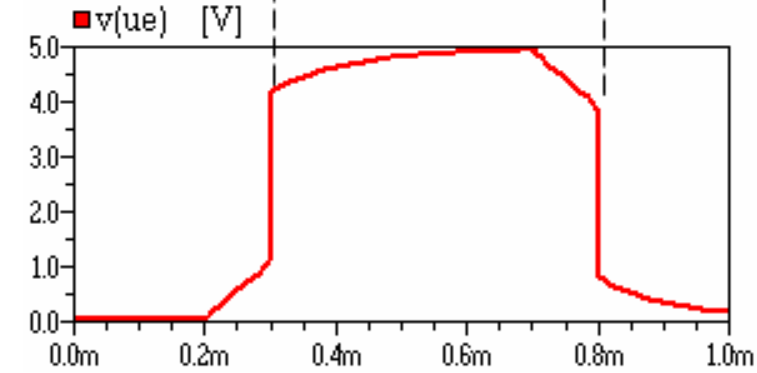
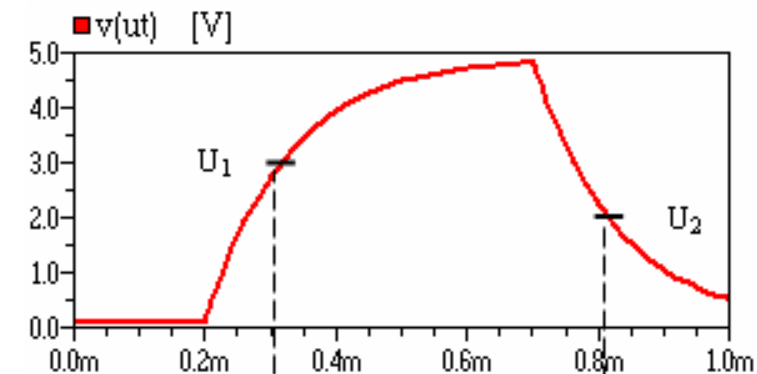
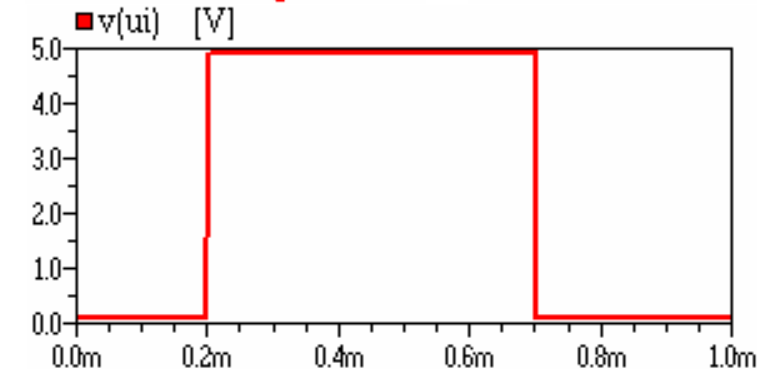
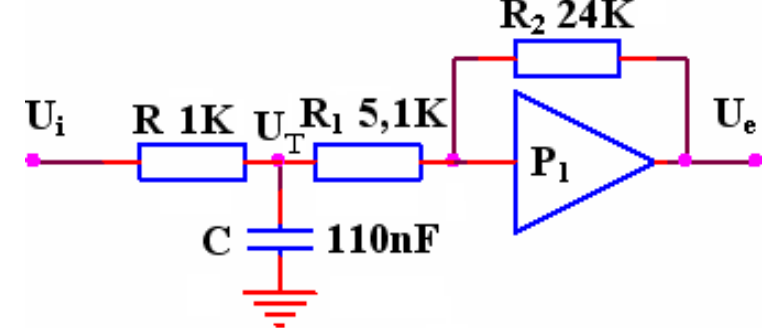
$$U_1 = \frac{R_1 + R_2}{2R_2} V_{DD} = 3V$$

$$U_2 = \frac{R_2 - R_1}{2R_2} V_{DD} = 2V$$

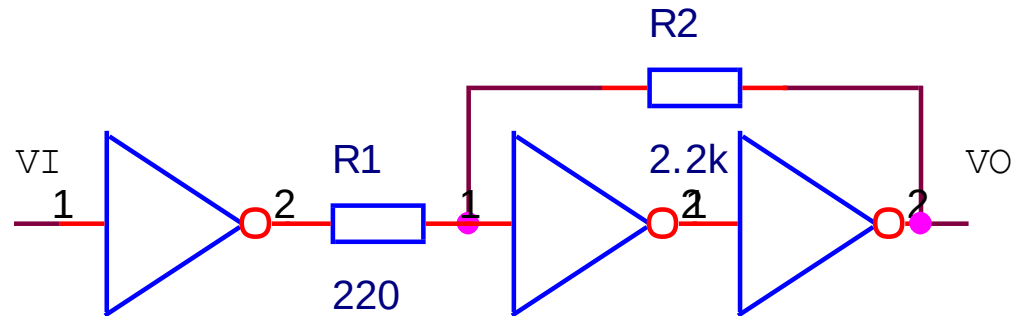
$$U_e(t) = U_e(\infty) + [U_e(0) - U_e(\infty)] e^{-\frac{t}{RC}}$$

$$C = \frac{t}{R \ln \frac{u_e(\infty) - u_e(0)}{u_e(\infty) - u_e(t)}}$$

$$C = \frac{t}{R \ln \frac{V_{DD}}{V_{DD} - U_1}} = \frac{t}{R \ln \frac{V_{DD}}{U_2}} = 110nF$$



Calculate the needed noise margins for a standard TTL gate driving a trigger Schmitt circuit built on TTL technology, with $R_1=220\Omega$ and $R_2=2,2k\Omega$.



$$U_1 = U_T \frac{(R_1 + R_2)}{R_2} - U_{OL} \frac{R_1}{R_2} = 1.4V \quad U_2 = U_T \frac{R_1 + R_2}{R_2} - U_{OH} \frac{R_1}{R_2} = 0.7V$$

$$M_H = V_{OH\min} - U_2 = 2.4V - 0.7V = 1.7V$$

$$M_L = U_1 - V_{OL\max} = 1.4V - 0.4V = 1V$$

- Design a delay circuit using TS circuit with TTL gates, implementing a delay of the positive edge with 0,1 ms. What's the delay for the negative edge?
- Design an oscillator with a TS circuit using CMOS gates. Oscillator is driven by a zero-active signal; oscillation period is of 2ms. CMOS gates are supplied with a power supply source of 5V.